

Sistem

$$\left. \begin{array}{l} y_1' = f_1(x, y_1, \dots, y_n) \\ \vdots \\ y_n' = f_n(x, \dots, y_n) \end{array} \right\} \vec{y}' = f(x, \vec{y})$$

če f_j linearne: linearni sistem

$$\Rightarrow \vec{y}' = A\vec{y}, A = A(x) \in \mathbb{R}^{n \times n}$$

Rešitve: $\vec{y} = c_1 \vec{k}_1 + \dots + c_n \vec{k}_n$; $c_j \in \mathbb{R}$, $\vec{k}_j$ lin. neodv. funke.

če $\vec{v}$ last. vekt. matrike A , je $\vec{y} = e^{\lambda x} \vec{v}$ rešitev en.

$$\begin{array}{l} y' = 4y - 3z \\ z' = 3y + 4z \end{array} \quad \begin{array}{l} \begin{bmatrix} y \\ z \end{bmatrix}' = \underbrace{\begin{bmatrix} 4 & -3 \\ 3 & 4 \end{bmatrix}}_A \begin{bmatrix} y \\ z \end{bmatrix} \end{array}$$

$$p_A(\lambda) = (\lambda - 4)^2 - (-9) = (\lambda - 4 + 3i)(\lambda - 4 - 3i)$$

$$\lambda = 4 \pm 3i$$

$$1) \lambda = 4 - 3i$$

$$A - \lambda I = \begin{bmatrix} 3i & -3 \\ 3 & 3i \end{bmatrix} \rightarrow \begin{bmatrix} i & -1 \\ 1 & i \end{bmatrix} \rightarrow \begin{array}{l} xi - y = 0 \\ x + yi = 0 \end{array} \rightarrow \vec{v}_1 = \begin{bmatrix} i \\ 1 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} y \\ z \end{bmatrix} = e^{(4-3i)x} \begin{bmatrix} i \\ 1 \end{bmatrix} = (e^{4x} \cos 3x + i e^{4x} \sin 3x) \begin{bmatrix} i \\ 1 \end{bmatrix} = \underbrace{\begin{bmatrix} -e^{4x} \sin 3x \\ e^{4x} \cos 3x \end{bmatrix}}_{w_1(x)} + i \underbrace{\begin{bmatrix} e^{4x} \cos 3x \\ e^{4x} \sin 3x \end{bmatrix}}_{w_2(x)}$$

$\rightarrow$ Baza prostora $\mathbb{R}$ reš. je $\{w_1, w_2\}$

$$\rightarrow \text{spl. reš. je } \begin{bmatrix} y \\ z \end{bmatrix} = c_1 e^{4x} \begin{bmatrix} -\sin 3x \\ \cos 3x \end{bmatrix} + c_2 e^{4x} \begin{bmatrix} \cos 3x \\ \sin 3x \end{bmatrix}$$

Če se A ne da diag., poleg lastnih potrebujemo še korenske vektorje:

$$\vec{v}_{m+r} \xrightarrow{A-\lambda I} \vec{v}_{m+r-1} \xrightarrow{A-\lambda I} \dots \xrightarrow{A-\lambda I} \vec{v}_{m+1} \xrightarrow{A-\lambda I} 0$$

korenski vekt.
last. vekt.

Rešitve:

$$e^{\lambda x} \vec{v}_{m+1}$$

$$e^{\lambda x} (\vec{v}_{m+2} + x \vec{v}_{m+1})$$

$$\vdots$$

$$e^{\lambda x} \left(\vec{v}_{m+r} + \dots + \frac{x^{r-1}}{(r-1)!} \vec{v}_{m+1} \right)$$

$$(A - \lambda I)^r \vec{v}_{m+r} = 0$$

- $r=1$: last. vekt.
- $r>1$: korenski vekt.

$$y' = -2y + z$$

$$z' = -y - 4z$$

$$A = \begin{bmatrix} -2 & 1 \\ -1 & -4 \end{bmatrix}$$

$$p_A(\lambda) = (-2-\lambda)(-4-\lambda) + 1 = \lambda^2 + 6\lambda + 9 = (\lambda+3)^2$$

$$\lambda = -3$$

$$\begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix} \begin{bmatrix} y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\rightarrow \vec{v}_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix} \text{ last. vekt.}$$

$$(A+3I)^2 = \begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \rightarrow w = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\rightarrow \tilde{v} = (A+3I)w = \begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix} \text{ last. vekt.} \checkmark$$

rešitvi:

$$\vec{y}_1 = e^{-3x} \tilde{v}$$

$$\rightarrow \text{spl. reš. } \vec{y} = c_1 \vec{y}_1 + c_2 \vec{y}_2$$

$$\vec{y}_2 = e^{-3x} (w + x \tilde{v})$$

$$\underline{\underline{\begin{bmatrix} y \\ z \end{bmatrix} = c_1 e^{-3x} \begin{bmatrix} 1 \\ -1 \end{bmatrix} + c_2 e^{-3x} \begin{bmatrix} x+1 \\ -x \end{bmatrix}}}$$

$$y' = y + 2z + u$$

$$z' = -8y - 9z - 3u$$

$$u' = 12y + 12z + 3u$$

$$A = \begin{bmatrix} 1 & 2 & 1 \\ -8 & -9 & -3 \\ 12 & 12 & 3 \end{bmatrix}$$

$$\begin{aligned} p_A(\lambda) &= (1-\lambda)(-9-\lambda)(3-\lambda) + 36(1-\lambda) + 72 + 16(3-\lambda) \\ &= -108 - 12(-9-\lambda) = \dots = -\lambda^3 - 5\lambda^2 - 7\lambda - 3 \\ &= -(\lambda+1)^2(\lambda+3) \end{aligned}$$

$$\lambda_1 = -1$$

$$\lambda_2 = -3$$

$$\begin{array}{c|c|c|c|c} & 1 & 5 & 7 & 3 \\ \hline -1 & & -1 & -4 & -3 \\ \hline & 1 & 4 & 3 & 0 \end{array}$$

λ_1 :

$$\begin{bmatrix} 2 & 2 & 1 \\ -8 & -8 & -3 \\ 12 & 12 & 4 \end{bmatrix} \rightarrow \vec{v}_1 = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$$

$$\begin{aligned} 2(x+y)+z &= 0 \\ 3(x+y)+z &= 0 \end{aligned}$$

$$(A - \lambda_1 I)^2 = \begin{bmatrix} 0 & 0 & 0 \\ 12 & 12 & 4 \\ -24 & -24 & -8 \end{bmatrix}$$

$$3(x+y)+z=0$$

λ_2 :

$$\begin{bmatrix} 4 & 2 & 1 \\ -8 & -6 & -3 \\ 12 & 12 & 6 \end{bmatrix} \rightarrow \vec{v}_2 = \begin{bmatrix} 0 \\ 1 \\ -2 \end{bmatrix}$$

$$x=0, z_y = -z$$

$$\rightarrow \vec{v}_3 = \begin{bmatrix} 1 \\ 0 \\ -3 \end{bmatrix} \rightarrow \vec{v}_1' = \begin{bmatrix} 2 & 2 & 1 \\ -8 & -8 & -3 \\ 12 & 12 & 4 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ -3 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$$

$$\Rightarrow \text{res.: } \begin{bmatrix} y \\ z \\ u \end{bmatrix} = c_1 e^{-x} \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} + c_2 e^{-3x} \begin{bmatrix} 0 \\ 1 \\ -2 \end{bmatrix} + c_3 \left(e^{-x} \begin{bmatrix} 1 \\ 0 \\ -3 \end{bmatrix} + x e^{-x} \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} \right)$$